

Zm00001eb375600 (*fbxl1*) — results of Analyses 0, 1, and 2, and a surprise: mapping bias inverts the reported direction of effect

To: Rubén Rellán-Álvarez **From:** Hannah Pil **Re:** Results from the follow-up memo of 2026-09-13

1. Summary

I ran Analyses 0, 1, and 2 from your memo on the BAMs we already had. The first-order findings match your framing; the second-order finding does not.

- Strand setting is correct.** The current pipeline's `strandSpecific = 1` is the winning setting by a clear margin. `-s 2` is only 9% assigned — the count matrix is not silently on the wrong strand. Analysis 0 is closed.
- Reference-mapping bias is real, genome-wide, and concentrates in the strongest cis hits.** At all 41,885 cis tests the distribution of β is nearly symmetric (51.4% teo-lowers). At $p < 1e-4$ it is 70% teo-lowers, and at $FDR < 0.05$ it is 63%. That signature — stronger filter, more skew — is the fingerprint of a technical rather than biological driver.
- fbxl1* is the extreme case: bias inverts the sign of the effect.** In the bias-exposed 3'UTR window, teo carriers recover 45% of B73's reads. In the bias-free terminal-CDS window, teo carriers have **3.08× more** reads than B73. The R_{div} / R_{cons} diagnostic is **0.147**. So the gene is not downregulated by teosinte introgression — it is upregulated, and the whole-gene DE result is a mapping artifact stronger than the biology it was masking.
- H3 (antisense-mediated repression) is not supported** by the aggregate window counts and can be closed without a per-sample analysis (§5).
- H2 needs to be reframed.** Your original H2 was "B73's ~5 kb intron-4 insertion is a positive element." Since teo is the higher-expressing allele, the sign of the H2 prediction reverses: **the B73 insertion is repressive, or disrupts a native positive element** that is intact in teosinte. That is the leading candidate now (§6).

I did not run Analysis 3 as a per-sample test — the reason is in §5.

2. Analysis 0 — strand setting is correct

Ran `Rsubread::featureCounts` on 5 BAMs (`PN1_SID1` through `PN1_SID13`) with `strandSpecific = 0, 1, 2`. Per-sample %Assigned:

Setting	Mean %Assigned
<code>-s 0</code> (unstranded)	71.4%

Setting	Mean %Assigned
<code>-s 1</code> (forward-stranded)	73.3%
<code>-s 2</code> (reverse-stranded)	9.4%

The gap between `-s 0` and `-s 1` is only ~2 pp (a strictly stranded library would give 20+ pp). BRB-seq R2 has enough ambiguous strand info that many reads read as either. Nothing needs to change; `-s 1` is the right call.

Files: `hannah/FBX_analyses/analysis0_strand_check/strandtest_s{0,1,2}.summary`.

3. Analysis 2 — genome-wide direction of cis effects

I plotted the distribution of MatrixEQTL cis β at four p-value thresholds (all tests, $p < 0.05$, $p < 1e-4$, $FDR < 0.05$):



Figure R1

Threshold	n tests	% teo-lowers
all tests	41,885	51.4%
$p < 0.05$	9,697	58.9%
$p < 1e-4$	3,230	70.3%
$FDR < 0.05$	6,471	62.6%

The pattern — nearly symmetric at the noise level, progressively more skewed as the filter gets stricter — is what you would predict from reference-mapping bias operating across the genome. Real biology exists in the “teo-raises” tail (PPL2 sits there at $\beta = +5.69$, the largest effect in the dataset), and the bias sits as a systematic layer on top of it. The concrete implication is that the SET of significant cis hits carries a bias signature that is not a per-hit judgment; each individual hit still needs to be evaluated on its own before being called biology.

I would report the significant-hits count with an explicit note about the skew rather than treating the number as face-value.

Files: `output/cis_eQTL/analysis2_bias_check/beta_direction_hist.png` and `beta_direction_summary.csv`.

4. Analysis 1 — R_div / R_cons for *fbxl1*

Ran your four windows through `featureCounts -s 1` on all 384 sample BAMs; computed per-window mean CPM by fbxl1 introgression status (39 teo carriers, 253 B73 background), then the memo’s diagnostic ratio.

Per-window Teo/B73 mean CPM ratio:

Window	mean CPM B73	mean CPM Teo	ratio Teo/B73
sense_terminalCDS_conserved (bias-free)	1.78	5.49	3.08
sense_3UTR_divergent (bias-exposed)	91.00	41.10	0.45
antisense_test_intron4	0.072	0.050	0.69
fiveprime_antisense_gene	0.517	0.494	0.96

R_div / R_cons = 0.147.

Figure R2 

Figure R3 

Reading of these numbers, against your §8 rubric.

Your rubric had four rows. The one that most closely matches this outcome is “< 1, effect persists in R_cons → H1 inflated a real effect; re-estimate from the conserved window.” That is essentially what happened, with one addition your rubric didn’t anticipate: R_cons is not just “the real downregulation, less extreme than reported” — R_cons is **greater than 1 and in the opposite direction from the gene-level result**. The bias didn’t just inflate a real down-regulation; it **inverted a real up-regulation** into an apparent down.

- The 3’UTR window carries roughly 50× more reads than the CDS window in either group. Whole-gene counts (which is what the A4 plot uses) are dominated by the 3’UTR signal.
- The 3’UTR window undersamples teo reads by roughly 7× relative to what the CDS ratio would predict (0.147 = the fold-drop between the two windows).
- Once that undersampling is accounted for, teo carriers actually express fbxl1 more than B73 lines.

Caveats I want to be transparent about.

- The CDS window is only 210 bp and receives few reads (median 1.25 CPM in B73, 2.66 in teo). The ratio 3.08 is a ratio of means and would swing if we picked a slightly different window or dropped low-depth samples. The *direction* (Teo > B73 in the CDS) is robust — visible in the per-sample boxplot as well as in the mean — but the exact fold is uncertain.
- Some teo carriers show 0 CPM in the CDS window (visible in the boxplot). Those are the same samples that show near-zero counts in the whole gene and probably have low library depth for this region rather than a real biological zero.

- I did not test whether $\text{Teo} > \text{B73}$ in the CDS window survives per-sample statistical inference (e.g., Wilcoxon, or LMM). At $n=39$ vs $n=253$ with a $\sim 2\times$ shift in a low-count window, a formal test is worth running before quoting a number. It is likely significant but I have not verified it.

Files: `output/FBX_analyses/analysis1_ratio/window_ratios_summary.csv`, `per_sample_window_cpm.png`, `ratio_by_window.png`.

5. On Analysis 3 (antisense) — closing without a per-sample test

Your memo made the per-sample antisense correlation the decisive test for H3. I would like to argue that the aggregate window counts already close H3:

- Mean CPM in the intron-4 antisense window is **0.072 in B73 vs 0.050 in Teo** — essentially at background in both, and slightly *lower* in teo carriers, opposite of the H3 prediction.
- H3 was proposed to explain a downregulation in teo. Since teo carriers are actually the higher-expressing group, there is no down-regulation for antisense to have caused.

I have not run the per-sample anti-correlation of antisense count against sense CPM. It is 10 minutes of code and I can do it if you would prefer a formally-tested null over an argument-based one. My current view: the memo's Question 1 ("present in carriers, absent in B73?") is already answered no, and Questions 2 and 3 were conditional on that being yes.

If we later see evidence that a small subset of teo carriers do carry antisense reads (per-sample, not on average), that reopens the question without contradicting the aggregate null.

6. Reframing H2 — the ~5 kb B73 intron-4 insertion is now the leading candidate

Your original H2 predicted "B73 insertion carries a positive element → B73 higher." Since we now see teo higher, the natural revision is:

H2'. The ~5 kb B73-lineage insertion in intron 4 is repressive of *fbx11* — either intrinsically (a TE-derived silencer, a splicing perturbation that lowers processed mRNA, or a length-driven decrease in transcription efficiency) or by disrupting a native cis-regulatory element that is intact in teosinte.

The evidence that fits this: - The CDS is 99.5–100% conserved, so it is not a protein change. - The intron-4 length is 7,965 bp in B73 versus 1,839–3,257 bp in the four teosintes. - Teo alleles (which lack the insertion) express higher; B73 alleles express lower.

The obvious first step is a TE annotation lookup on the B73 intron-4 insert. If it is a known LTR retrotransposon or MITE family, that names the mechanism directly. If it is a novel unique-sequence insert, we would need to look for regulatory-element predictions or (long-term) knock it out.

7. What this changes beyond *fbxl1*

Two things worth flagging that fall out of these results:

1. **Any cis effect calling teo-downregulation deserves an R_div/R_cons check before being trusted.** Your memo framed this as a single-gene diagnostic; the genome-wide skew in Analysis 2 shows the pattern is not confined to *fbxl1*. It is worth running the CDS-vs-3'UTR strategy on the top-N teo-lowers cis hits to see how many flip. Not a trivial exercise — requires per-gene window annotation — but the pipeline is now established.
 2. **Teo-raises cis hits (like PPL2 at $\beta = +5.69$) are relatively bias-safe.** Mapping bias can hide real up-regulation (as it did for *fbxl1*); it cannot manufacture apparent up-regulation from nothing. Effects in the teo-raises tail are the ones most defensible to report.
-

8. What I did not do

- **Per-sample statistical test on the CDS-window Teo > B73 result.** The effect is visible in the raw boxplot but I have not run a Wilcoxon or LMM to accompany the mean ratio.
 - **Analysis 3 as a per-sample test** (§5).
 - **TE annotation of the B73 intron-4 insertion.** Genome-side task, not BAM-side; blocked on none of the analyses in this memo.
 - **The genome-wide “which cis hits flip when you use the CDS trick?” scan.** This is the natural extension but is a larger analysis; I would want to scope it as its own memo.
 - **UMI-collapsed counting and TMM normalization** (your §7 notes). Cheap to add; unlikely to overturn any of the above. Worth doing before final numbers for a manuscript, not needed for this diagnostic pass.
-

9. Suggested next steps, in order

1. TE annotation of the B73 intron-4 insertion.
 2. Formal significance test on the CDS-window Teo > B73 effect (per-sample Wilcoxon).
 3. Decide whether to run the genome-wide R_div/R_cons scan for other teo-lowers hits. If yes, scope as a new memo.
 4. Decide how to talk about the A4 *fbxl1* result in the meantime — my current preference is to retire that panel from any talk material and replace it with the R_cons / R_div comparison if *fbxl1* has to be discussed at all.
 5. UMI dedup and TMM at the point where we commit to numbers.
-

Attachments

- [output/cis_eQTL/analysis2_bias_check/beta_direction_hist.png](#) (Figure R1)

- `output/FBX_analyses/analysis1_ratio/per_sample_window_cpm.png` (Figure R2)
- `output/FBX_analyses/analysis1_ratio/ratio_by_window.png` (Figure R3)
- `output/FBX_analyses/analysis1_ratio/window_ratios_summary.csv`
- `output/cis_eQTL/analysis2_bias_check/beta_direction_summary.csv`
- Scripts: `scripts/FBX_analysis0_strand_check.R`, `scripts/FBX_analysis1_count_windows.R`, `scripts/FBX_analysis1_ratio.R`